

Section 97 — Digit And Broken-Profile DP

Remember only the boundary between fixed and unknown choices

This lesson derives A–C. Problem D is editorial-only.

A. Tight digit DP

Process digits from most significant to least. `tight` says whether the chosen prefix still equals the bound; if false, every remaining digit may range through nine. For a digit-sum target, the other state is the accumulated sum. The base case accepts exactly the target.

To answer $[L, R]$, compute $\text{count}(R) - \text{count}(L-1)$. Leading zeroes are harmless for digit sums and naturally make zero one fixed-length representation. The state boundary is the prefix summary: the future depends on its exact digits only through `tight` and the sum.

B. Digit DP with an automaton

For a forbidden substring, compile the pattern with KMP failure links. The automaton state is the longest pattern prefix matching a suffix of the digits already emitted. A transition reaching the full pattern is rejected. Add `started` so padding zeroes are not mistaken for representation digits; handle the number zero explicitly at the base case.

This is a general product construction: digit-bound automaton times property automaton. The same method supports remainders, occurrence counts, and any finite-state language.

C. Broken-profile tiling

Scan one row while a mask records cells already occupied from the previous row. At the first free cell, every valid tiling makes exactly one local choice: a horizontal domino if its neighbor is free, or a vertical domino that sets a bit in the next-row mask. Blocked cells behave as already occupied but may never overlap incoming bits.

Induction on the first unprocessed cell proves that the recursion enumerates every tiling exactly once. The important ordering fact is that after a cell leaves the profile, no future domino can touch it.

Designing the digit state

Begin with the future question: given two equal-length prefixes, what facts must agree so that every possible suffix behaves identically? For a bounded digit sum these are **tight** and current sum. For a forbidden pattern, add the automaton suffix state. Add **started** only when leading zeroes must not be fed to the property automaton.

Prune sums that already exceed the target or cannot reach it with remaining digits. This changes speed, not the state proof. For a range, ensure $F(x)=0$ when $x<0$ so the formula also handles $L=0$.

A row-profile trace

At the start of a row, bit **c** says that cell (**row**,**c**) was occupied vertically from above. Scan to the first bit that is neither incoming nor blocked. A horizontal placement consumes **c**,**c+1** in the current row; a vertical placement sets bit **c** of the outgoing mask. When the row finishes, the outgoing mask is the complete information passed to the next row.

Never allow an incoming bit on a blocked cell, and ensure every current-row bit is consumed before advancing. Compare with exhaustive placement on boards up to a few dozen cells. Transposing the board so width is the smaller dimension often gives the decisive 2^{width} improvement.

Exercises

A–C build tight, automaton, and occupancy profiles. D transfers the row profile to local exclusion rather than covering.